

Correspondence: Node Reference to Node/Container Mapping

Source feature: Node Reference
Target feature(s): Node Mapping, Container Mapping
Automation type: FullAutomation - COMPLEX
Relation: GMF Node References act as indirection points that bind semantic containment to graphical realization via Node Mappings. Since Sirius distinguishes explicitly b etween simple nodes and containers, the migration must inspect the structural co ntext of the referenced Node Mapping. By conditionally creating a Container Mapp ing when compartments exist, or a Node Mapping otherwise, the mapping preserves both the containment semantics and the visual hierarchy encoded in GMF diagrams.

Main Strategy:

- 1. [A] take the node mapping (A) of the top node reference (B) or child reference (C)
- 2. [A] check if A has a compartment and a child reference (X)
- 3. [A] if X is true
 - 3.1. [A] create a container mapping under the default layer (for B) or under the parent r eference's equivalent (for C)
 - 3.2. [A] take all the compartment mappings (G) of A
 - 3.3. [A] take all the visual representation compartments (H) of G
 - 3.4. [A] take all the child accesses (I) using the figure and accessor properties of H
 - 3.5. [A] take all the shapes (J) using the figure property of I
 - 3.6. [A] take all the LayoutData features properties of J
 - 3.7. [A] execute "Computing Children Presentation for Container Mapping from Top Node/Chi ld Reference"
- 4. [A] else check if B/C has a sibling compartment (Y)
- 5. [A] if Y is true
 - 5.1. [A] take all the sibling compartment mappings (D)
 - 5.2. [A] take the compartment mapping in D that include B/C in the children property (E)
 - 5.3. [A] take E's equivalent container mapping (F)
 - 5.4. [A] create a (sub) node mapping under F

Strategies of subcorrespondences

Computing Domain Class

- 1. [A] copy the element property of A and paste it into domain class

Computing Semantic Candidates Expression

- 1. [A] copy the containment property of B/C and paste it into semantic candidates expre ssion

Computing Id

- 1. [A] copy the element property of A and paste it into id

Structural overview

